CONTENTS

7.1 INTRODUCTION 2

7.2 ASSESSMENT METHODOLOGY 3
Scope 3
Data sources 3
Assessment approach 3
Significance criteria 3

7.3 RELEVANT POLICY 6
National Planning Policy Framework (March 2012) 6
Swindon BSwindon Borough Local Plan 2011 (2006) 7
Draft Swindon Borough Local Plan 2026 7

7.4 BASELINE CONDITIONS 11
Noise Monitoring 11

7.5 POTENTIAL EFFECTS 13
Construction stage 13
Post-completion stage 18
NPPF/PPG24: Noise Assessment 20
BS:8233 Assessment of Day Time Noise Levels in Living Rooms 21
BS:8233 Assessment of Night Time Noise Levels in Living Rooms 22
Proposed School 22

7.6 MITIGATION MEASURES 24
Direct and indirect noise and vibration from construction 24
Direct façade noise levels on the proposed dwellings and proposed school 24

7.7 RESIDUAL EFFECTS 26
Summary of effects 26

7.1 INTRODUCTION

7.1.1 This chapter has been prepared by Brookbanks Consulting Ltd to present the findings of an assessment of the effects of the Proposed Development on noise and vibration during both the construction and post-completion stages.

7.1.2 Environmental noise rarely reaches the sound pressure levels associated with hearing impairment. However, noise can cause annoyance; it is commonly blamed for sleep disturbance and has been linked by researchers to less obvious effects, such as cardiovascular and mental health problems and reduced performance at work or school.

7.1.3 Human subjects, under laboratory conditions, are generally only capable of noticing changes in steady noise levels of no less than 3 dB(A).

7.1.4 The Proposed Development has the potential for noise impacts associated with operational traffic and railway movements; and also during the construction phase.

7.1.5 The following sections outline the Site conditions and assess the appropriateness of the Site for the Proposed Development in accordance with local and national guidance.

7.1.6 A glossary of common noise terminology is provided in **Appendix 7.1**.

7.2 ASSESSMENT METHODOLOGY

Scope

7.2.1 For the purposes of this assessment, the Study Area has been taken as the Application Site for the agreed NPPF/PPG24 assessment. A wider study to assess the impact within the local road network has been based on the Study Area adopted within the Transport Assessment.

7.2.2 Discussions have been held with Gupti Gosine and Damon Green, Environmental Health Officers at Swindon Borough Council (SBC), who discussed and agreed the scope of the assessment. Through these discussions, it was agreed that a noise survey is required for the Site due to sensitive receptors being identified. Vibration from the railway line has not been anticipated as having a significant impact on the Proposed Development and therefore vibration monitoring is not required.

Data sources

7.2.3 Gupti Gosine and Damon Green, Environmental Health Officers at Swindon Borough Council have been consulted throughout the assessment process and the preparation of this chapter.

7.2.4 This included an agreement that PPG24 and British Standard 8233:1999 being based on the 'Calculation of Road Traffic Noise' are the most appropriate methods to assess the noise environment.

Assessment approach

7.2.5 Methods of assessment have been employed that are consistent with current guidance and recommendations in the form of statutory documents and recognised publications to ensure that the findings are robust.

Significance criteria

7.2.6 The DMRB Volume 11, Section 3, Part 7: Environmental Assessment Procedure is used for the assessment of operational noise impacts for road schemes and gives guidance on the magnitude of impact from noise changes upon the local environment. The significance of predicted increases in road traffic noise as a result of the Proposed Development has been assessed according to the criteria described below.

7.2.7 Table 7.1 and 7.2 below outline the criteria for determining the magnitude of changes in traffic noise and Table 7.3 outlines the criteria for determining the magnitude of changes in Ambient Noise.

Table 7.1: Magnitude of Effect Short Term

Magnitude	Change in Traffic Noise (dB)
Large	5 +
Moderate	3 – 4.9
Small	1 – 2.9
Negligible	0.1 – 0.9

Table 7.2: Magnitude of Effect Long Term

Magnitude	Change in Traffic Noise (dB)
Large	10 +
Moderate	5 – 9.9
Small	3 – 4.9
Negligible	0.1 – 2.9

Table 7.3: Magnitude of Effect Short Term

Magnitude	Change in Ambient Noise (dB)
Large	5+
Negligible	0 – 4.9

Table 7.4 Sensitivity of Receptors

Sensitivity	Description
High	Dwellings
Medium	Schools, hospitals, quiet recreation areas, places of worship
Low	Offices, cafes/bars with external areas
Negligible	Industrial, retail

7.2.8 BS5228: 'Code of Practice for noise and vibration control on construction and open Sites' is the methodology for the prediction of construction noise, and control of noise and vibration. Significance can be considered in relation to fixed limits for noise and vibration, or alternatively in considering the potential change in the ambient noise level with the addition of construction noise for the purposes of the Proposed Development. This significance can be assessed using the criteria below.

Table 7.5: Significance of Effect

Magnitude	Sensitivity			
	High	**Medium**	**Low**	**Negligible**
Large	Major	Major	Moderate	Minor
Moderate	Major	Moderate	Minor	Negligible
Small	Moderate	Minor	Minor	Negligible
Negligible	Minor	Negligible	Negligible	Negligible

7.3 RELEVANT POLICY

National Planning Policy Framework (March 2012)

7.3.1 A Planning Policy Guidance 24 (PPG 24)[1] was introduced by the former Department of the Environment in 1994. PPG 24 provided advice on how the planning system can be used to minimise the adverse impact of noise without placing unnecessary restrictions on development or unduly adding to the costs and administrative burdens of business. PPG 24 contained advice to local authorities regarding the use of their planning powers to minimise the adverse impacts of noise when considering planning applications for new residential developments.

7.3.2 The Department for Communities and Local Government published the National Planning Policy Framework (NPPF)[2] in March 2012. This was produced to support the reforms of the planning system and to promote sustainable growth. The NPPF supersedes PPG24. However, in the absence of more up-to-date appraisal criteria, the Noise Exposure Category (NEC) criteria provided by PPG24 are largely accepted as the preferred methodology.

7.3.3 Through discussions with SBC's Environmental Health Officers, it has been agreed that the Noise Exposure Categories (NECs) for residential development set out within PPG24 should be used for the purpose of this assessment.

7.3.4 PPG 24 recommended appropriate levels for exposure to different noise sources, as indicated below.

[1] DOE PPG24: Planning and Noise (1994)
[2] Communities and Local Government NPPF (2012)

Table 7.6: PPG24 NEC Boundaries

NEC Boundary	Mixed Noise Sources		Planning Advice
	Daytime (07:00 – 23:00) $L_{Aeq\ 16hr}$ dB	Night Time (23:00 – 07:00) $L_{Aeq\ 8hr}$ dB	
A	<55	<45	Noise need not be considered as a determining factor in granting planning permission, although noise at the high end of the category should not be regarded as a desirable level.
B	55 – 63	45-57	Noise should be taken into account when determining planning applications and, where appropriate, conditions imposed to ensure an adequate level of protection against noise.
C	63 – 72	57-66	Planning permission should not normally be granted. Where it is considered that permission should be given, for example because there are no quieter sites available, conditions should be imposed to ensure a commensurate level of protection against noise.
D	>72	66>	Planning permission should normally be refused.

Swindon Borough Local Plan 2011 (2006)

7.3.5 Policy DS6 sets out standards of design and amenity to ensure development proposals are "compatible with, and protect the amenity of, nearby land uses in terms of visual intrusion, privacy, **noise disturbance**, smell, pollution (including light pollution), and safety."

Draft Swindon Borough Local Plan 2026

7.3.6 Policy EN7: Pollution states that:

> "Development that is likely to lead to emissions of pollutants such as **noise**, light, **vibration**, smell, fumes, smoke, soot, ash, dust, grit or toxic substances that may adversely affect existing development and vulnerable wildlife habitats, shall only be permitted where such emissions are controlled to a point where there is no significant loss of amenity for existing land uses, or habitats"

7.3.7 Further to this, Policy TR1: Sustainable Transport Networks indicates how Swindon's sense of place and quality of life can be supported and contributed towards by "Minimising the impact of congestion, **noise** and air quality."

.

British Standard 8233:1999; Sound Insulation and Noise Reduction for Buildings

7.3.8 BS8233:1999[3] gives recommendations for the control of noise in and around buildings and suggests appropriate criteria and internal noise limits for habitable rooms of residential dwellings. In accordance with the requirements of BS8233, the following internal and daytime noise limits will need to be met within sensitive rooms of the residential dwellings:

i. 30dB LAeq (16 hour) during the daytime in living rooms.
ii. 30dB LAeq (8 hour) during the night time in bedroom areas.
iii. 45dB LAmax should not be exceeded during the night time in bedroom areas.

Building Bulletin 93: Acoustic Design of Schools

7.3.9 BB93[4] provides guidance on external and internal noise levels to be achieved at school development sites. The following daytime noise levels should be achieved:

- An upper limit of 60dB LAeq (30 minutes) at the boundary of external premises for teaching and recreation.
- 55 dB LAeq (30 minutes) in unoccupied playgrounds, playing fields and other outdoor areas.
- 50 dB LAeq (30 minutes) in at least one area of unoccupied playgrounds, playing fields and other outdoor areas, to ensure suitable noise levels for outdoor teaching.
- Indoor ambient noise limits in schools between 30 and 40 dB LAeq (30 minutes) depending on the use of the room.

[3] British Standard 8233:1999; Sound Insulation and Noise Reduction for Buildings
[4] DeFS (2003) Building Bulletin 93: Acoustic Design of Schools

Calculation of Road Traffic Noise

7.3.10 The Calculation of Road Traffic Noise[5] is the standard UK procedure for defining measurement and calculation methods for assessing road traffic noise. The procedures assume typical traffic and noise propagation conditions which are consistent with moderately adverse wind speeds and direction during the specified periods.

7.3.11 All predicted noise levels are expressed in terms of LA10 (1-hour) or LA10 (18-hour) between 0600 and 2400. The LA10 (18-hour)) is the arithmetic average of the values of LA10 hourly data for each of the eighteen 1-hour periods between 0600 and 2400.

Calculation of Rail Noise

7.3.12 The Calculation of Rail Noise[6] document provides the method of calculating railway traffic noise levels for new and altered railways. The document is referred to as the method of assessment to be used in the Noise Insulation Regulations. This guidance can also be used for carrying out railway noise assessments for planning purposes.

7.3.13 The calculation takes into account noise generation from rail traffic sources such as rail/wheel interface and diesel power sources. The screening effects of any barriers including ground levels and buildings are assessed alongside distance attenuation. Reflection effects are also taken into account and the combined noise level at a given point is calculated for all the different train types passing the receptor.

British Standard 5228: 'Code of Practice for Noise and Vibration Control on Construction and Open Sites'

7.3.14 BS5228: 'Code of Practice for noise and vibration control on construction and open sites'[7] is the methodology for the prediction of construction noise, and control of noise and vibration. This Standard provides guidance on methods of predicting and

[5] Department of Transport (1988) The Calculation of Road Traffic Noise
[6] Department of Transport (1995) The Calculation of Rail Noise
[7] British Standard 5228: 'Code of Practice for Noise and Vibration Control on Construction and Open Sites'

measuring noise and assessing its impact on those exposed to it, and also recommendations for basic methods of vibration control.

7.3.15 Suitable methods are provided for the calculation of noise from construction activities, including basic information regarding noise levels from a range of construction equipment.

7.3.16 The standard provides guidance for the identification of the significance of noise and vibration levels from surface construction activity. Significance can be considered in relation to fixed limits for noise and vibration, or alternatively in considering the potential change in the ambient noise level with the addition of construction noise.

7.3.17 There are no national noise criteria for limiting noise from construction sites. BS 5228 Annex E gives different methods of guidance on the significance of noise effects from construction and recommends the ABC method to establish construction noise limits.

7.3.18 The ABC method involves rounding the existing ambient noise levels to the nearest 5dB for the appropriate time period and then comparing these levels to the total noise level, including construction noise. If the total noise level exceeds the existing rounded value, then a significant effect is deemed to have occurred.

Table 7.7: Thresholds of significant effect from construction noise

Assessment Period	Category A Threshold	Category B Threshold	Category C Threshold
Night time (23:00 to 07:00)	45	50	55
Evenings and weekends	55	60	65
Day time (07:00 to 19:00) and Saturdays (07:00 to 13:00)	65	70	75

7.3.19 If the existing ambient noise levels exceed the threshold values presented above then a significant effect is deemed to have occurred. The ABC method should only apply to residential receptors.

7.4 BASELINE CONDITIONS

Noise Monitoring

7.4.1 Existing Noise measurements have been carried out adjacent to the roads on the boundary and running through the Site, opposite the railway line to the south and alongside the industrial area to the west. The results have been used to validate the 3D noise mapping produced by SoundPLAN (**Appendix 7.2**).

7.4.2 Daytime and night time noise levels have been monitored over a 24 hour period, together with manned recording to ensure that any noise sources from the employment areas were identified.

7.4.3 All acoustic measurement equipment used during the noise surveys conformed to Type 1 specification of British Standard 61672: 2003: Electroacoustics, Part 1 Specifications.

Table 7.8: Noise Monitoring Equipment Inventory

Equipment Description	Manufacturer	Serial number
Sound Level Meter	**Norsonic 118**	28952
Sound Level Meter	**Norsonic 118**	30559
Sound Level Meter	**Norsonic 118**	30560
Acoustic Calibrator	**Norsonic 1251**	32856

7.4.4 Traffic noise along the A420 was identified as the most significant source of noise, with railway noise also identified as significant. No other significant noise sources have been recorded. The results of the survey are indicated below together with the locations of the noise monitoring.

Table 7.9: Noise Monitoring Results

Location	Daytime LAeq 16hr	Daytime LAmax 16hr	Night time LAeq 8hr	Night time LAmax 8hr
Location 1	**63**	66	**57**	61
Location 2	**67**	71	**62**	66
Location 3	**60**	64	**55**	59

Location 4	**66**	68	**58**	62
Location 5	**54**	59	**45**	49

7.4.5 The predominant noise source for this area is the traffic noise from the A420 and the railway. The results of the noise survey indicate that the Site boundary lies predominantly in NEC B/C with the majority of the Site lying in NEC A. The noise survey demonstrated that traffic and rail noise are the predominant noise sources.

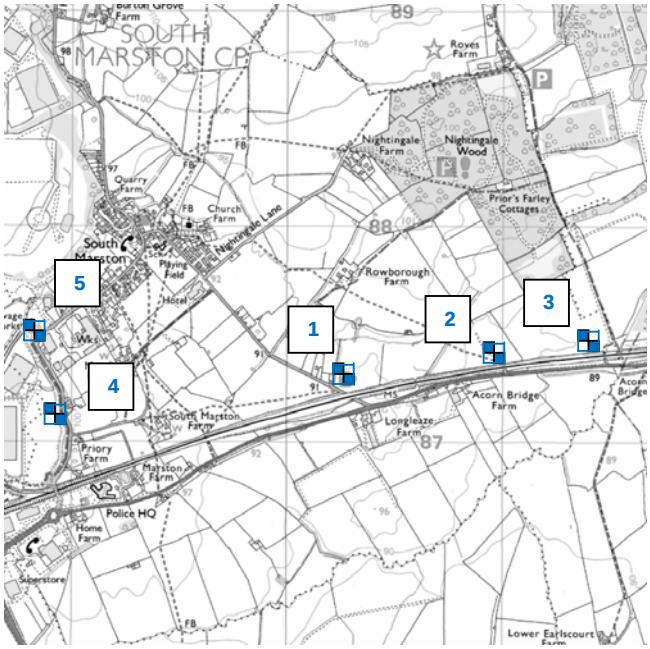

Figure 7.9: Noise Monitoring Locations

7.5 POTENTIAL EFFECTS

Construction stage

7.5.1 During the construction stage, it is envisaged that limited demolition, landform creation, installation of necessary services and building construction will form the main noise impacts upon existing residential properties in the environs of the Site.

7.5.2 Although the final details of the construction activities cannot be finalised until construction contractors have been confirmed, the noise levels have been predicted using the sound pressure levels for the plant as described in BS 5228: 2009 Part 1. The sound pressure levels in BS 5228 have been presented as a LAeq at a distance of 10 m. A high percentage for the 'on-time' (the length of time that the equipment remains active on site) has been assumed so as to present a possible worst case.

Table 7.10: List of Construction Plant and Associated Sound Levels

Plant Description	BS5228 Reference	Sound level at 10m	On time %
Angle Grinder	Table C4 No. 93	80	40
Asphalt Paver	Table C5 No. 33	75	60
Circular Saw	Table C4 No. 72	79	40
Compressor	Table C5 No. 5	75	80
Concrete Pump & Concrete mixer truck discharging	Table C4 No. 28	79	80
Concrete Saw	Table C4 No. 71	85	10
Delivery Lorry	Table C2 No. 35	80	70
Diesel Generator	Table C4 No. 84	74	100
Dozer	Table C5 No. 12	77	60
Dumpers	Table C4 No. 9	77	60
Excavator	Table C5 No. 34	82	75
Percussion Drill	Table C4 No. 69	85	40
Pneumatic Breaker	Table D2 No.2	81	40
Poker Vibrator	Table C4 No. 33	78	80
Road Planer	Table C5 No. 7	82	70
Roller Compactor	Table C5 No. 29	76	60

Telescopic Handler	Table C4 No. 54	79	75
Tower Crane	Table C4 No. 49	77	60
Tracked Excavator	Table C5 No. 18	80	70
Tracked Excavator fitted with Breaker	Table D2 No. 5	91	70
Tracked Mobile Crane	Table C4 No. 52	75	60
Vibratory Roller (22t)	Table C5 No. 28	77	60
Water Pump	Table C2 No. 45	65	75
Welder	Table C3 No. 31	73	40
Wheeled Loader	Table C2 No. 26	79	75

7.5.3 The on-time correction factor has been extracted from Figure F5 within BS5288.

7.5.4 The construction noise impacts have been calculated using the following formula as described in BS5228:

$$k_h = 20 \times LOG \frac{R}{r}$$

Where:

K_h = the correction for propagation across hard ground

R = the distance to the receptor location

r = the distance of 10 m at which the SPL has been measured

7.5.5 Where more than one piece of the same equipment is used in a construction activity, the following equation has been used to determine the total noise level generated:

Combined noise level = $x + 10*\log_{10}(N)$

Where

x = noise level from a single piece

N = the number of items of equipment used

7.5.6 To calculate the combined noise level for a construction process the following equation has been used to combine the noise levels from the individual construction plant:

Combined event= $10*\log_{10}(10^{L1/10} + 10^{L2/10} + 10^{L3/10} + ...10^{Ln/10})$
Where L_1 = individual noise event

7.5.7 A possible worst case has been presented by considering propagation across hard ground and by not considering screening provided by topographical features, buildings or other structures.

7.5.8 The potential noise impacts during the construction stage are presented below.

Table 7.11: Demolition Noise Levels

Plant	Number	Noise level at 10m	Noise level at 20m	Noise level at 50m	Noise level at 100m	Noise level at 200m
Pneumatic breaker	1	77.0	71.0	63.0	57.0	51.0
Tracked excavator with fitted breaker	1	89.5	83.5	75.5	69.5	63.5
Concrete Saw	1	75.0	69.0	61.0	55.0	49.0
Dumper	1	78.0	72.0	63.0	58.0	52.0
Diesel generator	1	74.0	68.0	60.0	54.0	48.0
Total		90.2	84.3	76.2	70.3	64.3

Table 7.12: Site Mobilisation Noise Levels

Plant	Number	Noise level at 10m	Noise level at 20m	Noise level at 50m	Noise level at 100m	Noise level at 200m
Delivery Lorry	1	78.5	72.5	64.5	58.5	52.5
Tracked Mobile Crane	1	73.0	67.0	59.0	53.0	47.0
Telescopic Handler	1	78.0	72.0	64.0	58.0	52.0
Wheeled loader	1	77.5	71.5	63.5	57.5	51.5
Dozer	1	75.0	69.0	61.0	55.0	49.0

Dumpers	2	78.0	72.0	63.0	58.0	52.0
Diesel generator	1	74.0	68.0	60.0	54.0	48.0
Total		85.2	79.2	71.0	65.2	59.2

Table 7.13: Road Construction Noise levels

Plant	Number	Noise level at 10m	Noise level at 20m	Noise level at 50m	Noise level at 100m	Noise level at 200m
Road Planer	1	80.5	74.5	66.5	60.5	54.5
Tracked Excavator	1	78.5	72.5	64.5	58.5	52.5
Dozer (Spreading fill)	1	75.0	69.0	61.0	55.0	49.0
Dumpers	2	78.0	72.0	63.0	58.0	52.0
Vibratory Roller (22t)	1	75.0	69.0	61.0	55.0	49.0
Asphalt Paver	1	73.0	67.0	59.0	53.0	47.0
Diesel Generator	1	74.0	68.0	60.0	54.0	48.0
Total		85.5	79.5	71.3	65.5	59.5

Table 7.14: Site Clearance Noise Levels

Plant	Number	Noise level at 10m	Noise level at 20m	Noise level at 50m	Noise level at 100m	Noise level at 200m
Dumpers	2	78.0	72.0	63.0	58.0	52.0
Tracked Excavator	1	78.5	72.5	64.5	58.5	52.5
Lorry	1	78.5	72.5	64.5	58.5	52.5
Dozer	1	75.0	69.0	61.0	55.0	49.0
Compressor	2	77.0	71.0	63.0	57.0	51.0
Diesel Generator	1	74.0	68.0	60.0	54.0	48.0
Total		84.9	78.9	70.8	64.9	58.9

Table 7.15: Building Construction Noise Levels

Plant	Number	Noise level at 10m	Noise level at 20m	Noise level at 50m	Noise level at 100m	Noise level at 200m
Tracked Excavator	1	78.5	72.5	64.5	58.5	52.5
Diesel Generator	1	74.0	68.0	60.0	54.0	48.0
Dumpers	1	75.0	69.0	61.0	55.0	49.0
Telescopic Handler	1	78.0	72.0	64.0	58.0	52.0
Concrete Pump & Concrete mixer truck discharging	1	78.0	72.0	64.0	58.0	52.0
Poker Vibrator	2	80.0	74.0	66.0	60.0	54.0
Compressor	2	77.0	71.0	63.0	57.0	51.0
Total		86.1	80.1	72.1	66.1	60.1

7.5.9 Construction activities produce significantly high noise levels, particularly close to source. Construction noise tends to fluctuate and is usually of fairly short duration. The construction noise impacts will depend on the proximity of construction activities to nearby receptor locations.

7.5.10 The demolition and construction noise impacts predicted above indicate that the highest impacts occur within 200m of the Site. The predicted noise levels are based on a possible worst case scenario. Propagation across hard ground has been assumed and no screening from topographical features or other structures has been assumed. The majority of existing residential dwellings lie over 200m from the centre of the site meaning the highest value identified for noise levels at 200m (maximum) is 64.3dB, which is below the Category A threshold (ABC method) of 65dB. However, as expected, construction will take place across the whole site, as a result further mitigation will be required in the form of the Construction Environmental Management Plan to minimise operations during sensitive time periods. With regards to the demolition of the outbuildings at Rowborough Farmhouse, this will be short term and is not considered to have a significant impact.

7.5.11 Therefore, given the nature of the construction activities the impact will not be significant in relation to the closest receptors and at most will have a minor adverse impact pre-mitigation.

Post-completion stage

Local network traffic noise impacts

7.5.12 Traffic noise predictions have been made using the CRTN prediction methodology. The methodology has been used to predict the magnitude of any change in noise level resulting from the development proposals at the roadside of the local network.

7.5.13 The predicted changes in noise level, identified with respect to the road traffic noise impact assessment criteria, are presented in Table 7.15 showing the 2026 with and without the completed development impact respectively.

Table 7.15: Predicted Noise Levels Within Local Road Network

Link	Noise Level without Proposed Development	Noise Level with Proposed development	Noise Impact	Magnitude
A313 Ocotal Way	69.5	70.0	0.5	Negligible
A419 Between Commonhead And Whitehart	77.3	77.4	0.1	Negligible
A419 between J15 and Commonhead	75.9	75.9	0.0	Negligible
A419 north of Whitehart	77.7	77.7	0.0	Negligible
A420 at Whitehart	74.6	75.7	1.1	Negligible
A420 east of Gablecross	73.1	74.2	1.1	Negligible
A420 east of Old Vicarage Lane	73.0	73.5	0.5	Negligible
A420 west of Gablecross	74.3	75.5	1.2	Negligible
A420 west of Old Vicarage Lane	72.9	73.9	1.0	Negligible
A4259 at Commonhead junction	75.0	75.1	0.1	Negligible
A4259 Queens Drive	72.8	72.6	-0.2	Negligible
A4312 Drakes Way	73.7	74.0	0.3	Negligible

A4312 Queens Drive	75.6	75.8	0.2	Negligible
B4006 Dorcan Way	70.9	71.0	0.1	Negligible
B4006 Swindon Road	74.1	73.9	-0.2	Negligible
Covingham Drive	71.3	71.6	0.3	Negligible
Ermin Street at Whitehart	68.7	68.9	0.1	Negligible
Fleming Way	71.9	72.1	0.2	Negligible
Highworth Road north of Old Vicarage Lane	66.3	65.4	-0.9	Negligible
Link between Honda RB and Hyde Rd	77.1	77.1	0.0	Negligible
Link between Hyde Road and Bypass	77.1	77.0	-0.1	Negligible
M4 Motorway	78.7	78.5	-0.2	Negligible
Marlow Avenue	67.8	67.9	0.1	Negligible
Merlin Way at Whitehart	72.8	73.3	0.5	Negligible
Old Vicarage Lane north of A420	59.2	66.9	7.7	Moderate
Old Vicarage Lane south of Highworth Road	59.1	60.7	1.6	Negligible
Oxford Road at Whitehart	71.2	71.8	0.6	Negligible
Pack Hill at Commonhead junction	62.0	62.2	0.2	Negligible
Purley Road at Commonhead junction	67.2	67.2	0.0	Negligible
Supermarket & truck stop access south of Gablecross	68.1	68.7	0.6	Negligible
Thornhill Road north of Gablecross	67.6	69.2	1.6	Negligible
Thornhill Road west of Highworth Road	66.4	64.9	-1.5	Negligible
Wanborough Road (east of A419)	68.2	68.3	0.1	Negligible

7.5.14 This demonstrates that the maximum increase that any of the traffic noise receptors are likely to experience as a result of the Proposed Development is 7.7dB, which has been identified as a moderate increase. This increase in noise is as expected as it takes place along Old Vicarage Lane north of the A420, which would alter in character from a relatively quiet rural route to principle access route into the Site. This is not anticipated to have a significant adverse impact due to no residential

dwellings existing along this section of Old Vicarage Lane. The remaining receptors will experience a negligible increase in noise levels. It is considered that an increase of less than 3db is not discernible and therefore it is concluded that the development generally will have a negligible impact.

NPPF/PPG24: Noise Assessment

7.5.15 Noise level prediction of the existing situation has taken place through computer modelling software SoundPLAN. The noise model incorporated recent elevation data that highlighted all the pertinent Site features. The assessment was based on recent traffic flows as provided by the Transport Consultant and railway flows based on the national rail timetables. The traffic flows were also provided for the future year assessment with and without the Proposed Development

7.5.16 The NEC boundaries for the existing situation were modelled initially, as indicated in drawings 1335-NM-01 and 1335-NM-02 in **Appendix 7.2**. This was repeated for the future baseline in 2026 as shown in 1335-NM-03 and 1335-NM-04. This indicated a close relationship between the modelled noise levels and those recorded through the noise survey.

7.5.17 The daytime and night time NEC boundaries for the Proposed Development were then modelled and indicated in 1335-NM-05 and 1335-NM-06. The resultant daytime and night-time noise contours indicate that the Site mainly falls into NEC A with the southern boundary falling into NEC B and C with a narrow corridor closest to the railway within NEC D. Until a detailed design of the Proposed Development has emerged, it is difficult to comment on the impacts on the dwellings. However, it can be assumed that properties will be set back from the railway line, with façades facing the railway line and A420 which would potentially lie partially within NEC C.

7.5.18 PPG24 advises that for sites that fall into NEC A:

> "Noise need not be considered as a determining factor in granting planning permission, although noise at the high end of the category should not be regarded as a desirable level"

7.5.19 PPG24 advises that for sites that fall into NEC B:

"Noise should be taken into account when determining planning applications and, where appropriate, conditions imposed to ensure an adequate level of protection against noise."

7.5.20 PPG24 advises that for sites that fall into NEC C:

"Planning permission should not normally be granted. Where it is considered that permission should be given, for example because there are no quieter sites available, conditions should be imposed to ensure a commensurate level of protection against noise."

7.5.21 Properties falling within NEC B/C typically require nothing more than standard thermal double glazing to provide an acceptable noise environment. Table 1 in Annex 6 of PPG24 provides examples of typical noise reductions for a dwelling façade with windows set in a brick/block wall. The table shows various levels of noise reduction provided by different glazing configurations and for different noise sources. The values shown are the level difference (in dBA) between the outside and the inside of a typical dwelling and to represent a worst case, it is assumed that the outside level is a façade measurement. For a road traffic noise spectrum, PPG 24 states that standard thermal double glazing will provide a façade sound insulation performance of 33 dB(A).

7.5.22 However, opening windows for ventilation purposes would reduce the insulation provided by the building façade and may cause internal design standards to be exceeded. Therefore, if it is considered necessary to satisfy internal noise standards with a degree of ventilation, mitigation measures may be required to enable occupiers to obtain ventilation with windows closed. Passive acoustic ventilators with the appropriate attenuation can be installed within the walls of habitable rooms or utilise 'trickle' vents within the window frames.

7.5.23 The layout of the Proposed Development and the internal arrangements of properties will be subject to further detailed design. Before the consideration of double glazing and trickle vents, priority should be given to the internal layout of the properties such that sensitive areas, i.e. bedrooms, are located to avoid facing onto the railway line and primary routes directly and consideration should be given to orientating buildings to minimise windows that face onto the noise source.

BS:8233 Assessment of Day Time Noise Levels in Living Rooms

7.5.24 BS8233 indicates that a daytime noise level of 30 dB LAeq represents a "good" standard and 40 dB LAeq as "reasonable". The calculated noise levels have been used to determine likely noise levels and the extent of attenuation required.

7.5.25 The actual location of housing within the Proposed Development would be determined at detailed design stage. Therefore a potential housing location fronting the railway and A420 has been selected. This indicates that the daytime façade noise level is 66.1 dB. This noise level reduces to 33.1 dB when taking into account noise reductions through thermal double glazing, which represents a "reasonable" internal noise standard.

7.5.26 This indicates that appropriate attenuation can be achievable for all of the properties through the use of thermal double glazing, with facades of properties further into the Site being protected and screened by other buildings. Orientating properties and considering the internal layout to avoid direct sight lines onto the main roads will further mitigate and reduce internal noise sources.

BS:8233 Assessment of Night Time Noise Levels in Living Rooms

7.5.27 BS8233 indicates that a night-time noise level of 30 dB LAeq represents a "good" standard in bedrooms and 35 dB as "reasonable". The calculated noise levels have been used to determine likely noise levels and the extent of attenuation required.

7.5.28 A typical night time façade noise level is identified as 62.4 dB. This noise level reduces to 29.4 dB when taking into account noise reductions through thermal double glazing, which represents a "good" internal noise standard.

7.5.29 This indicates that appropriate noise attenuation will be achievable, with facades of properties further into the Site being protected and screened by other buildings.

Proposed School

7.5.30 The dominant source of noise at the proposed school site will be from traffic. The Development Framework Plan does not provide the level of detail on the orientation of the school building and its internal or external layout for a detailed noise

assessment to be undertaken with reference to BB93. However, the school has been assumed as a single storey building(s) with parking areas, hard and soft play areas and sports playing fields.

7.5.31 The results of the noise modelling indicate that the school site is located in NEC A (<55 dB). A receptor located on the potential school building edge and a free field receptor located in the centre of the school site indicates a noise level of 51.3dB and 50.7 dB respectively.

7.5.32 This indicates that the BB93 upper limit at the boundary could be achieved and the internal classroom standard is also met, when considering the inclusion of thermal double glazing.

7.5.33 The noise levels across the school site will be dependent upon the detailed layout of the school, the locations of the playing fields and the use of the rooms. Depending on orientation, the school building could provide noise attenuation benefits to the playground areas.

7.6 MITIGATION MEASURES

Direct and indirect noise and vibration from construction

7.6.1 To minimise the impact on receptors during the construction process, the following generic noise and vibration mitigation measures need to be implemented as appropriate for all works and would be incorporated into the future Construction Environmental Management Plan (CEMP):

- Construction activities should be confined to times of the day when they are least likely to be disturbing;
- Careful selection of plant, construction methods and programming. Only plant conforming with relevant national or international standards, directives and recommendations on noise and vibration emissions should be used;
- Construction plant should be located, as far as is reasonably practicable, away from adjacent occupied buildings or as close as possible to noise barriers or site hoardings where these are located between the plant and the buildings;
- Static and semi-static plant/equipment (e.g. compressors and generators) should be fitted with suitable enclosures where practicable;
- Personnel will be instructed on best practice to reduce noise and vibration as part of their induction training and as required prior to specific work activities;
- When plant is not being used, it should be shut down and not left to idle;
- Methods of work and vehicular routes will be selected with regard to minimising noise and vibration impact;
- Given the phasing of construction, certain areas of the Proposed Development will be occupied while construction is still underway in adjacent areas. Where possible, the occupancy of completed phases of construction should be planned in such a way that there is a buffer between occupied areas and areas where construction is being carried out.

7.6.2 Given the nature of the construction activities expected on site, the impact could be significant without mitigation. However the construction noise and vibration impacts can be mitigated effectively through the CEMP.

Direct façade noise levels on the proposed dwellings and proposed school

7.6.3 Increases in traffic can increase noise levels at sensitive receptors, as detailed in Section 7.5. However, those increases can be kept within acceptable limits through the following mitigation measures:

- Passive ventilation systems and double glazing for only those residential properties falling within NEC C and fronting onto the railway line and A420.
- Internal layout of properties to consider the location of lounge and bedroom areas for those properties fronting onto the railway line and A420; and
- Site layout to consider the orientation of residential buildings to reduce sight lines onto the railway Line and A420.

7.7 RESIDUAL EFFECTS

7.7.1 The assessments completed above have considered both the proposed dwellings and the existing residential dwellings. It has been noted via the CRTN method of assessment that the existing dwellings along Old Vicarage Lane will experience a negligible increase in noise levels, highlighting no significant adverse impacts. The dwelling at Rowborough Farm will experience demolition of the surrounding outbuildings, however, this will be short term and is not considered to have a significant long term impact. As a whole, the assessments do not identify any significant adverse impacts and thus no residual effects are anticipated.

Summary of effects

7.7.2 The effects identified are summarised in Table 7.16 below:

Table 7.16: Summary of effects

Potential effect	Significance (pre-mitigation)	Mitigation measure	Significance of residual effect
Construction stage			
Construction/ Demolition Noise and Vibration	**Minor Adverse**	Application of the CEMP	**Negligible**
Post-completion stage			

Noise levels at the proposed dwellings and school	**Moderate Adverse**	Passive ventilation systems and double glazing for only those residential properties falling within NEC C and fronting onto the railway line and A420. Internal layout of properties to consider the location of lounge and bedroom areas for those properties fronting onto the railway line and A420; and Site layout to consider the orientation of residential buildings to reduce sight lines onto the railway Line and A420.	**Negligible**